

TRIASHA SARKAR

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PROFESSIONAL SUMMARY

First-year PhD student at Georgia Tech with research experience in sustainable transportation systems, lifecycle emissions modeling, applied statistics, and data-driven modeling. My work connects aerospace systems research with broader questions of transportation sustainability, mobility resilience, accessibility, and the distribution of environmental costs and benefits across communities. Current projects span sustainable aviation fuel, Atlanta road-network resilience and accessibility modeling, and machine-learning analysis of complex engineering data. I am deepening my foundation in regression, statistical inference, uncertainty quantification, and data-driven modeling to support policy-relevant transportation and sustainability research.

RESEARCH INTERESTS

Sustainable transportation systems · Mobility resilience and accessibility · Environmental sustainability, emissions, and life-cycle analysis · Applied statistics and regression modeling · Data-driven modeling and uncertainty quantification · Distributional and equity dimensions of transportation systems

EDUCATION

- PhD in Aerospace Engineering** | Georgia Institute of Technology, Atlanta, GA May 2025 - Present
- Graduate Research Assistant, Aerospace Systems Design Laboratory (ASDL); research spanning sustainable transportation systems, life-cycle analysis, applied statistics, and machine learning for engineering systems.
 - Graduate coursework: ISYE 6739 Statistical Methods (A); ISYE 6414 Regression Analysis (in progress, Summer 2026).
- MSc in Aerospace Vehicle Design** | Cranfield University, UK Mar 2022 - Feb 2023
- First Class Honours; research on hybrid propulsion, liquid hydrogen, alternative fuels, and environmental noise mitigation.
- B.Tech in Aerospace Engineering** | SRM Institute of Science and Technology, India Jul 2017 - Jun 2021
- First Class Honours.

TECHNICAL SKILLS

Statistics & Methods: Regression and ANOVA, statistical inference, probability, hypothesis testing, uncertainty quantification, linear algebra, optimization foundations

Sustainability & Transportation Systems: Life-cycle assessment (LCA), techno-economic analysis (TEA), greenhouse-gas/emissions modeling, carbon-intensity analysis, accessibility and network-resilience modeling

Data Analysis & Machine Learning: Regression modeling, Gaussian Process Regression, time-series anomaly detection, clustering, feature extraction, neural networks, reduced-order modeling, Operator Inference

Programming & Tools: Python (scikit-learn, PyTorch, NumPy, SciPy, pyDOE2), NetworkX, OSMnx, GeoPandas, MATLAB, C++, Git/GitHub, Jupyter, NASTRAN, PATRAN, ANSYS

RESEARCH EXPERIENCE

- Sustainable Aviation Fuel Lifecycle & Techno-Economic Analysis (Project GREEN TEA)** | Georgia Tech ASDL / U.S. Endowment for Forestry Sep 2025 - Apr 2026
- Techno-economic and life-cycle assessment of sustainable aviation fuel from forest residue under parametric uncertainty; modeled energy pathways, feedstock logistics, greenhouse-gas emissions, carbon intensity, and uncertainty in conversion assumptions.
 - Benchmarked carbon-intensity estimates against established regulatory and life-cycle frameworks including GREET, CORSIA, and 40BSAF-GREET.
- Atlanta Mobility Resilience Digital Twin** | Independent Project 2026 - Present
- Building a prototype digital twin of Atlanta's road network to model disruption scenarios and compare travel-time and accessibility outcomes across origin-destination pairs.
 - Developing the project toward larger stress-test scenarios, access to essential services, and neighborhood-level equity/vulnerability layers.
- Alternative-Configuration Aircraft under Climate Targets (FAA ASCENT 64)** | Georgia Tech ASDL May 2025 - Apr 2026
- Physics-based parametric modeling of alternative long-range aircraft configurations under uncertain CO2 targets within the ICAO LTAG framework.
- Demand & Economic Modeling for a Lifting-Body Airship (Project EAGLE)** | Vehicle Grand Challenge: RIC Enterprise / Georgia Tech ASDL Sep 2025 - Apr 2026
- Parametric sizing and demand modeling for a 750+ passenger lifting-body airship, including fare-sensitivity and fleet-sizing analysis across uncertainty scenarios.
- ML for Failure Detection in Solid Rocket Motor Simulations** | Georgia Tech, Dr. Adam Cox (AE8900) May 2026 - Present
- Statistical anomaly-detection pipeline for silent simulation failures in thrust-time histories, combining time-series outlier detection, clustering, CNN-based feature extraction, and phase-aware preprocessing.
- Surrogate Model Learning: GP, RSM, and RBF Comparative Study** | github.com/triasha72/Surrogate-model-learning 2025 - Present
- Compared Gaussian Process, RSM, and RBF surrogates across benchmarks; studied how approximation assumptions affect error, uncertainty, and failure modes in nonlinear regimes.
- NURBS-BEM Electromagnetic Solver with Learned Surrogates** | github.com/triasha72/NURBS_BEM_EMSolver Oct 2025 - Present
- Built a mesh-free electromagnetic solver for solenoid geometries using NURBS and a boundary-element method; generated parametric datasets for surrogate modeling, POD feasibility analysis, and Operator Inference exploration.
- ANN Regression on Power-Plant Data** | Georgia Tech (AE6394) Oct 2025
- Built a feedforward neural network in PyTorch on the UCI Combined Cycle Power Plant dataset ($R^2 = 0.94$) with a Streamlit dashboard for interactive inference.
- Research Thesis: SSBJ Noise Mitigation** | Cranfield / SENECA (Rolls-Royce, DLR, ONERA) Dec 2022 - Feb 2023
- Primary-nozzle geometry optimization for FAA FAR 91.817 noise compliance in a data-scarce, simulation-expensive design space.

INDUSTRY EXPERIENCE

- Systems Engineer** | Alten India Pvt Ltd (embedded at Rolls-Royce), Bangalore Oct 2023 - Apr 2025
- Airworthiness certification of the Trent XWB-84 EP engine; model validation against certification standards, with direct exposure to consequences of model failure in deployed systems.

Summer Intern, Structures Bangalore Aircraft Industries (BAIL)	Jun 2019 - Jul 2019
• Structural stress and material analysis under environmental loading using NASTRAN and PATRAN.	
Summer Intern, Avionics STAR, Surat	Jun 2021 - Jul 2021
• Avionics design with Arduino for solid-rocket-motor static-test safety and data acquisition.	

CERTIFICATE

- Introduction to Artificial Intelligence - IBM / Coursera (Jun 2020)